

Homework 6

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Deadline: 21.03.2024(13:59 hours)

1 Problem 1 (50 Points)

Consider a housing data given in file *data.txt*. The price (target) of an apartment depends on size of an apartment.

1. Load and plot the data, and explain why the price could be modeled by a predictor of the form $\hat{y} = c_1 * y_1 + c_2$, where $\hat{y}$ is the predicted price of a house given its size y_1 . (10 points)
2. Use the closed form formula for least squares and determine the analytical solution for $c = [c_1, c_2]^T$. After obtaining c , verify the fit on the data with one plot. (10 points)
3. Find c numerically using gradient descent with constant step size. You might want to either standardize the data or use different stepsize for c_1 and c_2 . This is because the slope is much smaller than the bias term. Make a plot showcasing the convergence of c as a function of iteration number. Add an explanation on how you overcame the scale issue. (30 points)

2 Problem 2 (50 Points)

Consider the economic dispatch problem in figure 1.

1. Write the optimization problem in a clean form with one objective function and 3 constraints. The decision variable is $P = [P_1, P_2, P_3]^T$. (10 Points)
2. Solve the problem analytically by utilizing Lagrangian multipliers. You might want to solve the system of equations with software. (20 points)

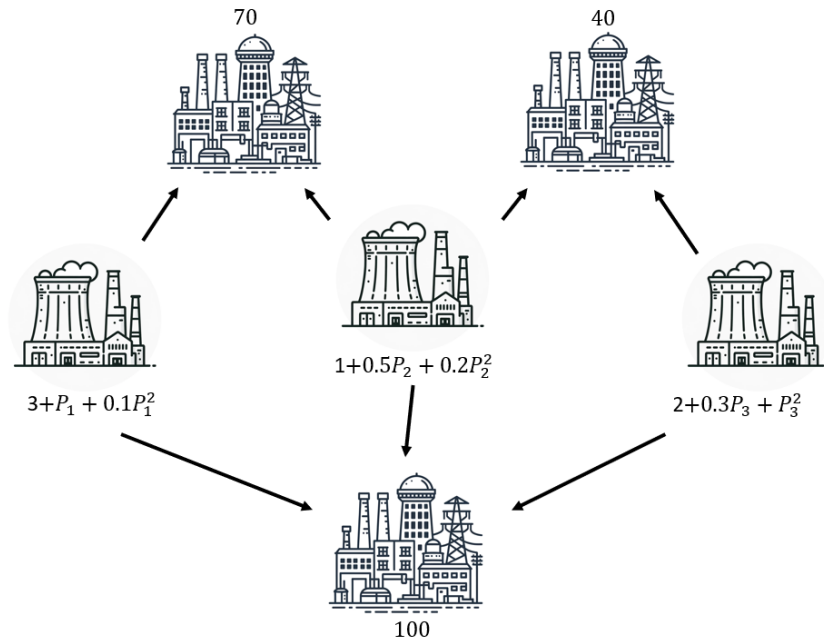


Figure 1: What are optimal P_1, P_2 and P_3 , given demand constraints?

- Solve the problem numerically using projected gradient descent. The problem should converge in 1 iteration. (20 points)